Divij Motwani

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Education

University of California, Berkeley Berkeley, CA
B.S. Electrical Engineering & Computer Sciences (EECS) *Expected May 2028*

Experience

Palantir Technologies New York, NY
Neurodivergent Fellow, Forward Deployed Engineering Intern *May 2026 – Present*

UC Santa Cruz — IR & Knowledge Management Lab Santa Cruz, CA
Student Researcher, Published Poster NeurIPS 2025 *Sep 2023 – August 2025*

- Co-authored **RAGUARD** fact-checking benchmark (Published NeurIPS 2025 (Main Conference), FEVER-8 2025) spanning 2,648 U.S. presidential-campaign claims (2000–2024) linked to 16,331 Reddit-sourced evidence docs labeled supporting / misleading / unrelated (avg. 6.2 docs/claim).
- Benchmarked seven LLM-RAG systems; misleading-only retrieval caused a mean 46.5% relative accuracy drop; GPT-4o most robust, Claude 3.5 strongest zero-context baseline. 33 Citations (Google Scholar)

OralAI — 1st Place, Regeneron ISEF, Top ∼40 of >10,000,000 students Palo Alto, CA
Co-founder, prev. in talks with Philips Sonicare *Jan 2023 – Present*

- Designed UV-fluorescence dental diagnostic system with custom 405 nm intra-oral camera and Raspberry Pi.
- Trained image-segmentation models: 98.4% precision (tooth detection), 93.4% (plaque segmentation).
- Deployed full-stack mobile app with real-time inference; cloud backend on AWS Cloud + MongoDB Atlas.
- **U.S. Patent #12632962**: *System and Method for Intraoral Imaging and Analysis for Preventing Dental Disease.*

Selected Projects

Period, end-to-end car-parking world model *2026*

- Built an 11M-parameter world model (based on DIAMOND's diffusion UNet) that parks a car in unseen lots, running at 120hz on a MacBook; trained from scratch on 7 hrs of general driving + 1 hr of task-specific trajectories (won comma hack 6).
- Added a 17K-parameter action head off the bottleneck to jointly predict curvature and acceleration alongside the diffusion denoising loss; dropped the image decoder at inference for ∼10x speedup with no accuracy loss; deployed live on a Tesla Model 3

FIRST Robotics Team 6036, Top 10/3200 worldwide *2023 – 2025*

- Veteran Build & Software member; developed high-throughput perception pipelines for control and tracking using fiducials.
- Adapted NASA JPL's **mrcal** calibration stack into a multi-threaded, multi-coprocessor architecture; integrated with custom autonomous control to achieve centimeter-level 3D pose estimation relative to the field.
- Prototyped and fabricated hardware with CAD/CAM for rapid iteration; optimized weight/stress; maintained motor and battery management systems; executed real-time repairs as pit crew using precision fabrication tools.

LEO Satellite Network Simulator *2025*

- Developed a software control and simulation system for LEO satellite mesh networks, modeling optical inter-satellite laser links and geostationary exit nodes with persistent ground-based laser backhaul.
- Simulated GPS-independent star-tracker localization for network entry, improving low-cost access for high-speed orbital networking.
- Built an interactive CesiumJS dashboard to visualize latency, network resilience, collision avoidance, and smart-contract-based allocation policies.

Zeta — AI Research Discovery System *AI Agent Hackathon (Exa × Anthropic × AWS × Lightspeed) 2025*

- Built a large-scale automated research system: crawled and parsed 6,117 recent *arXiv* Computation & Language papers (1.2M sentences); used Claude Opus to classify and locate 32,000 future-work statements and latent sentiments.
- Embedded and clustered statements with BGE-Large; performed graph analysis to identify cross-disciplinary research gaps, rank hypotheses, and surface high-leverage interdisciplinary directions.
- Created *ZetaEvolve*, an AlphaEvolve-inspired genetic-algorithm engine that converts hypotheses into fast microexperiments.

Honors & Awards

Regeneron ISEF Grand Award: 1st Place Biomedical Engineering (2024)

Optiver Opticode Cipher Challenge at NeurIPS '25: First Place Winner (2025)

UC Berkeley Citadel Securities Trading Challenge: First Place Winner (2025)

Susquehanna Discovery Program: Invited to SIG's discovery program, offered Summer '27 Quant Trading Internship (2026)

MIT CRE[AT]E Assistive Technology Competition: Winner (2024)

Pfizer DRL Global AI Flight Competition Finalist: Top 3 contestants internationally (Nominated for Grand Prize) (2025)

Stanford BASES Challenge: Round 2 Qualifier (top 20 of 200+ startups) (2025)

U.S. National Gallery for Young Inventors: 1 of 6 selected nationwide (2024)

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, Swift

ML/AI: PyTorch, TensorFlow, CUDA, Transformers, YOLO, Weights & Biases

Tools: Docker, Git, Linux, Pandas, Scikit-Learn, Xcode, OpenCV, Conda, React, Next.js, Node.js, Vercel, Solidity, Noir

Hobby: Classically trained tenor singer, v5 rock climber @ Berkeley Mosaic, film & digital photography, print & digital journalism